



STATISTICAL INFERENCE & LEARNING

Predicting **Excessive** Enforcement Patterns in Police Behavior After **Vehicles Stops in New York**

An empirical evaluation of post-stop arrest patterns, non-linear dynamics, and model robustness
in New York City (2023–2025)

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Data Intro, Data Pipeline & Population Split



0. Data Origin

The data comes from the **NYPD Vehicle Stop Reports**, published by the City of New York



1. Inspection of Raw NYPD Data

2,488,293 initial stop records (2023–2025) with
19 raw attributes covering time, coordinates,
officer command, and post-stop outcomes



2. Preprocessing & Cleaning

Parsing temporal components, explicit grouping,
turning of missing demographics into *Unknown*,
command aggregation into borough areas...



3. Dataset Subsetting

Separating vehicles to isolate two populations:
2-Wheeled (2W) vehicles and private cars (4W)

ⓘ Strategic Methodological Choice: Split Analyses

Keeping together both 2-wheeled and 4-wheeled observations hides the sample-specific distribution of the data, so we can 2 separate analyses with the same objective

Two Populations, Two Problems: Rare and Imbalanced Response

Univariate EDA, a first look

As soon as we separated the populations, we immediately noticed important differences with respect to the distribution of the predictors and response

2-Wheeled Vehicles (2W)

189,610 STOPS

5.6% Arrest Rate

- ✓ Total Arrests: 10,560
- ✓ Highly seasonal profile (peaks in warm months)
- ✓ Younger motorist demographic distribution
- ✓ Pronounced geographic disparities across boroughs

4-Wheeled Vehicles (4W)

2,157,445 STOPS

3.3% Arrest Rate

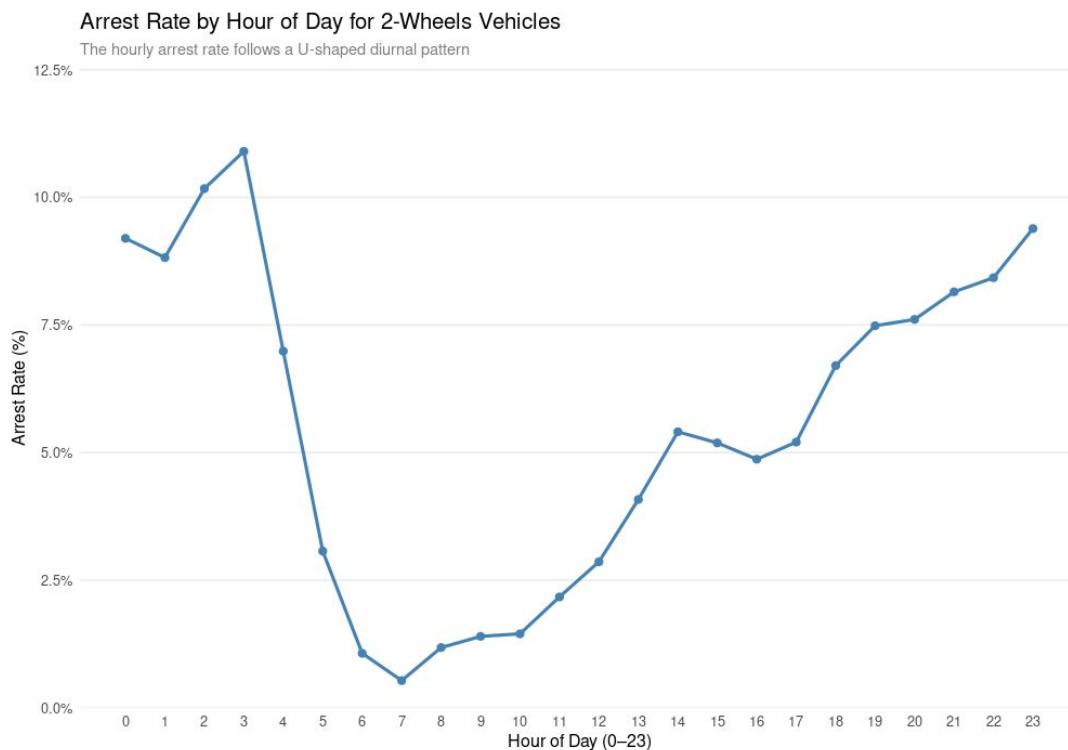
- ✓ Total Arrests: 70,600
- ✓ Stable year-round volume profile
- ✓ Broader, more mature age distribution
- ✓ Extrema rare outcome challenge for modeling

EDA: Similar Hour Rate Pattern

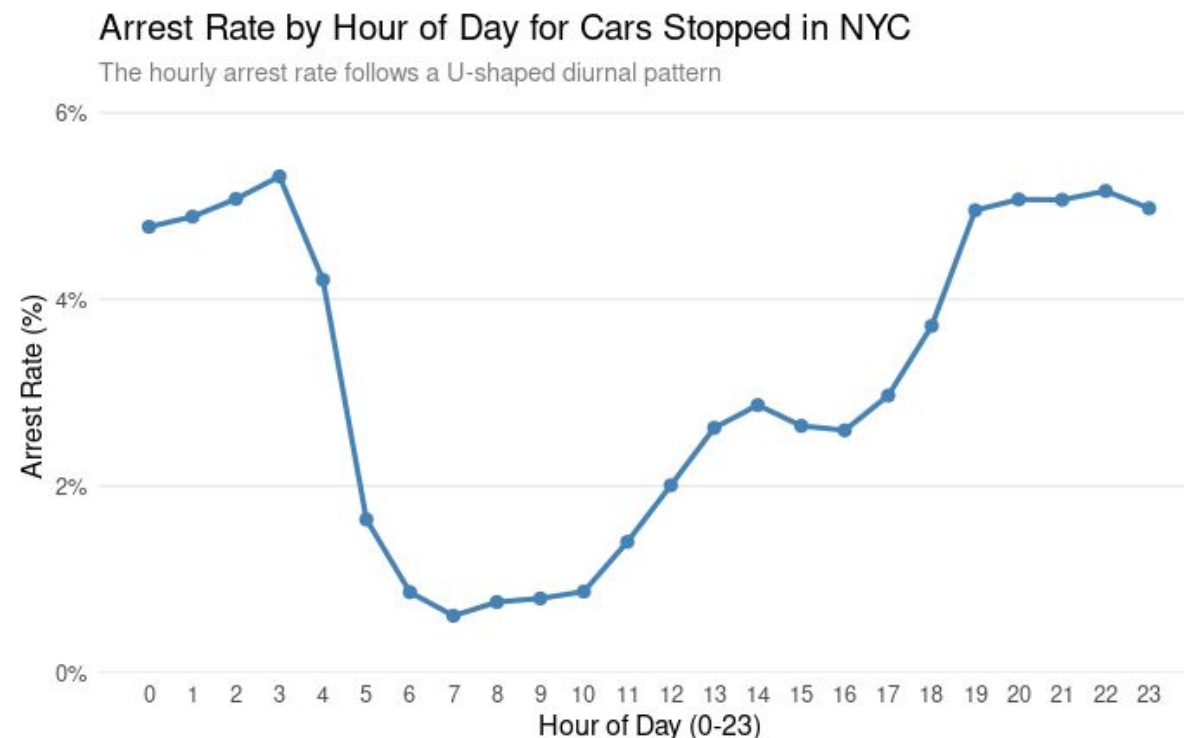
Diurnal Arrest Rate Trajectory

The raw arrest rate immediately show a non-linear 24-hour cycle across both populations, which suggests the use of splines in GAM

2W Hour vs Arrest



4W Hour vs Arrest

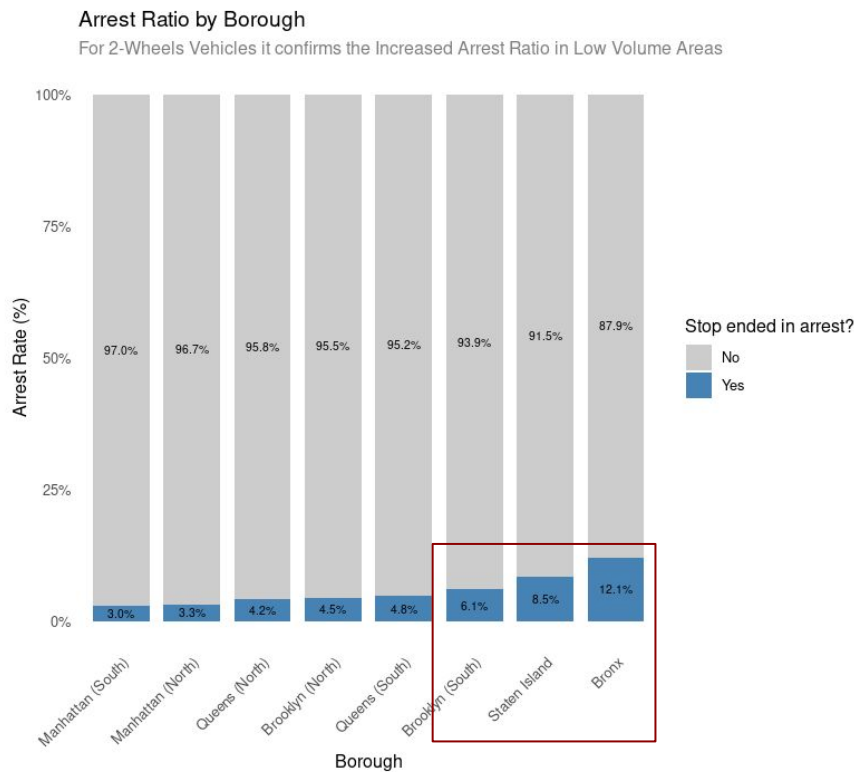


EDA: Geographic Disparities

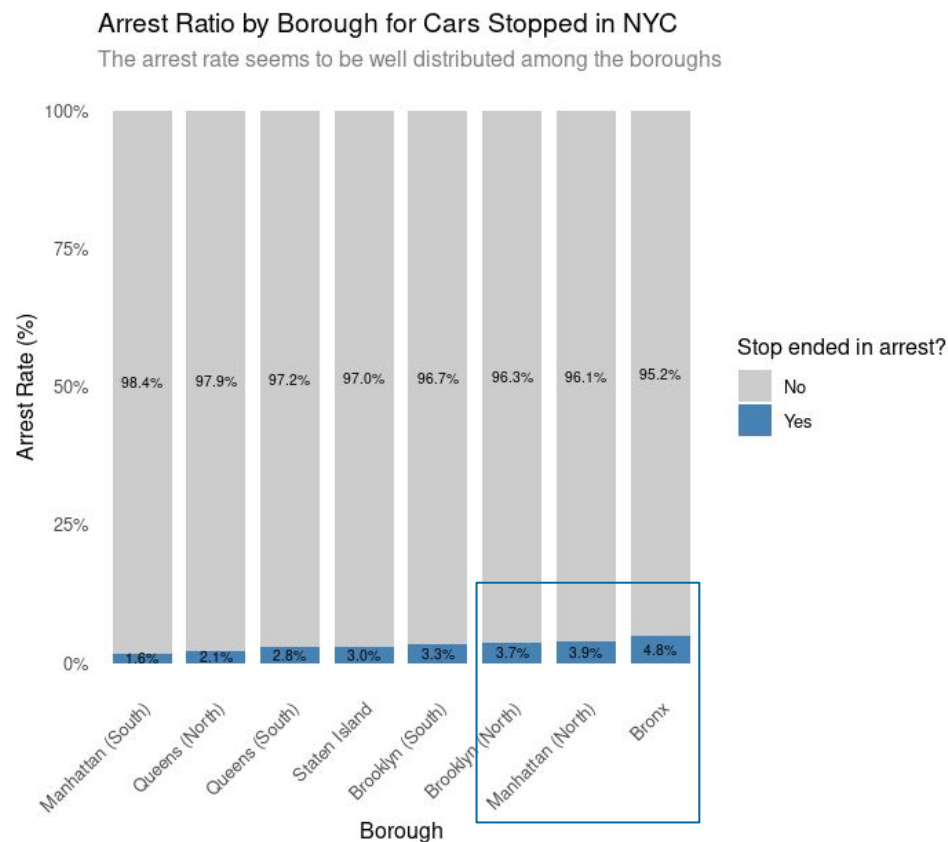
Borough Context

Raw post-stop arrest rate is far from uniform across NYC boroughs, with the Bronx consistently exhibiting the highest rates

2W Boro vs Arrest



4W Boro vs Arrest

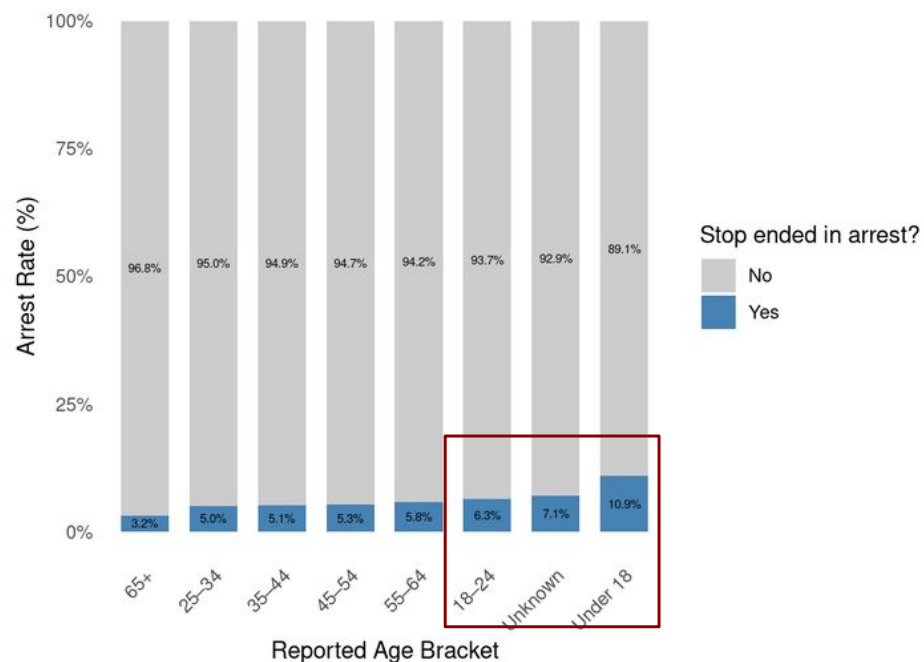


EDA: Age Rates Disagree

2W Age vs Arrest

Arrest rates are highest among the youngest motorists (<18)

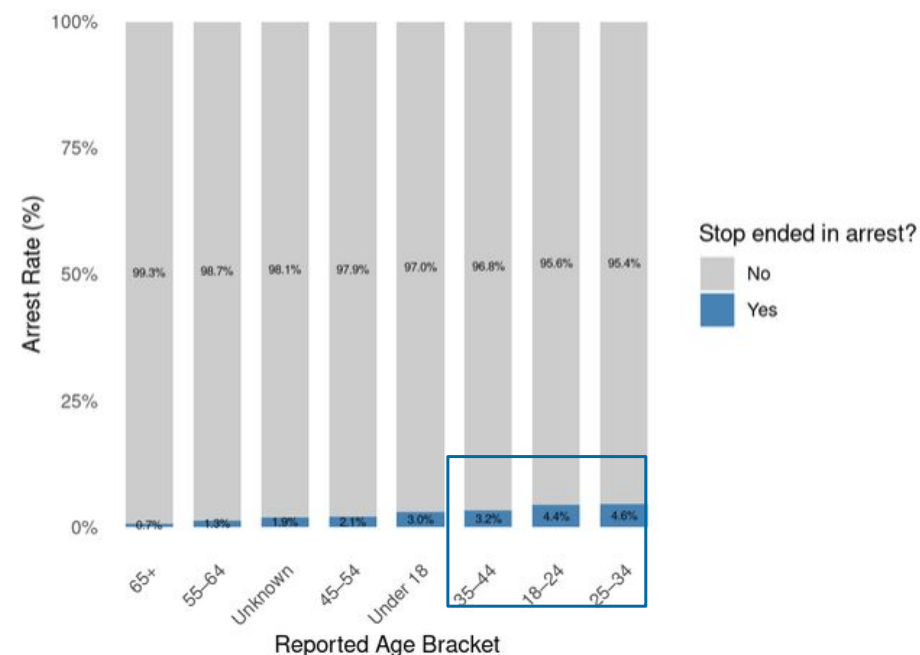
Arrest Ratio by Reported Age Bracket for 2-Wheels Vehicles Stopped
The 16-24 age bracket and Unknown show the highest arrest rates



4W Age vs Arrest

Arrest rates peak in the 18-24 and 25-34 age brackets, followed by a continuous decline for older motorists

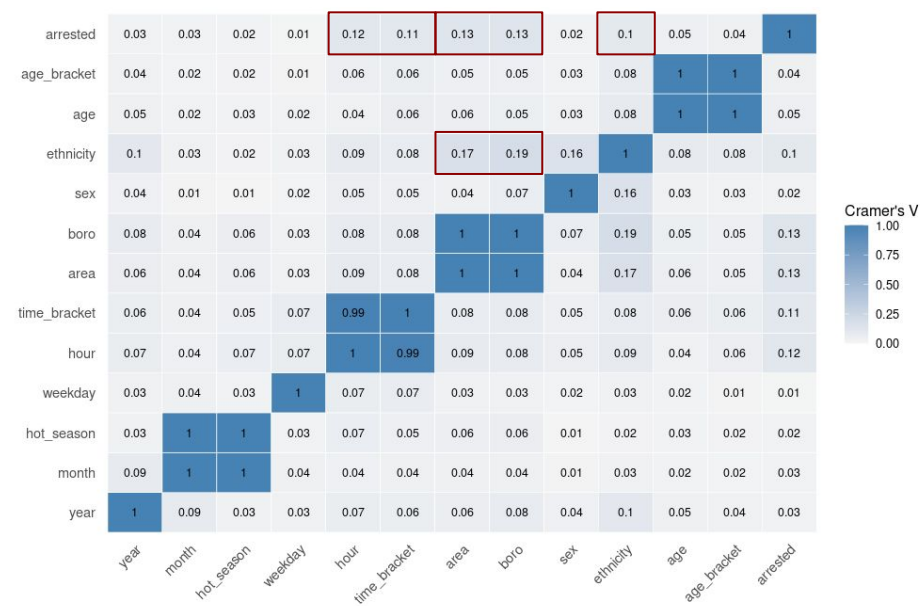
Arrest Ratio by Reported Age Bracket for Cars Stopped in NYC
The 18-34 age bracket shows the highest arrest rates



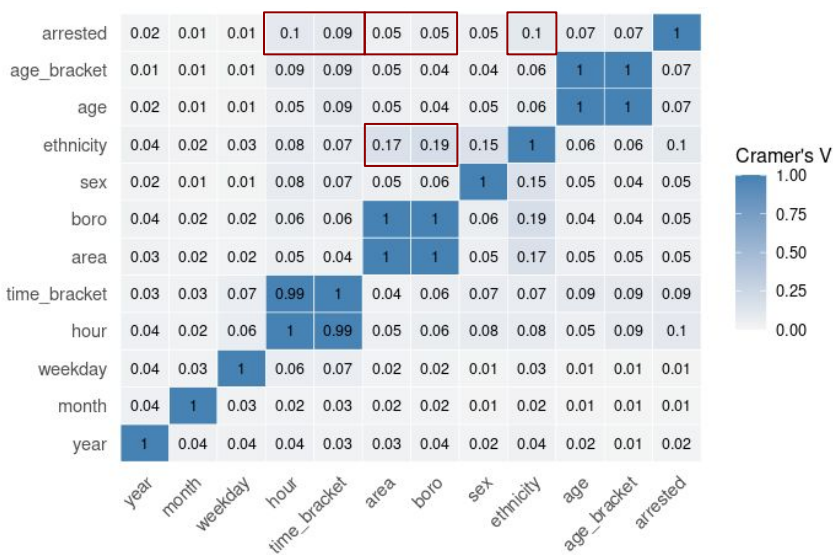
Cramér's V Predictor Matrix



2-Wheeled Vehicles (2W) – Cramer's V Association Matrix



4-Wheeled Vehicles (4W) – Cramer's V Association Matrix



Pairwise Associations

Cramér's V matrices indicate that individual bivariate associations with arrest are moderate

- Geography (Area/Boro): $V \approx 0.13$ (2W) | $V \approx 0.06$ (4W)
- Time (Hour/Bracket): $V \approx 0.11$ (2W) | $V \approx 0.09$ (4W)
- Ethnicity: $V \approx 0.10$ (2W) | $V \approx 0.10$ (4W)
- Predictor-Predictor: Highest between Ethnicity and Geography ($V \approx 0.17-0.19$).

Core Takeaway from Bivariate Analysis

"No single predictor independently (linearly) explains post-stop arrest decisions. However, models should focus on temporal, spatial, and demographic information."

Four Modeling Perspectives

Model Formulation

At the end of EDA and older analyses (not included) we came to the following fully additive formulation of the model (where all predictors are factors)

```
arrested ~ year + month + weekday + hour + boro + sex + ethnicity + age_bracket
```

Modelling Logic

While EDA allows us to find candidate predictors, here we aim to use the models in order to answer the following questions



1. Logistic

Question: What remains visible after adjustment for the other available predictors?



2. LASSO

Question: Does aggressive L1 shrinkage erase the primary signals?



3. GAM

Question: Are key numerical predictors (hour, age) non-linear?



4. Naive Bayes

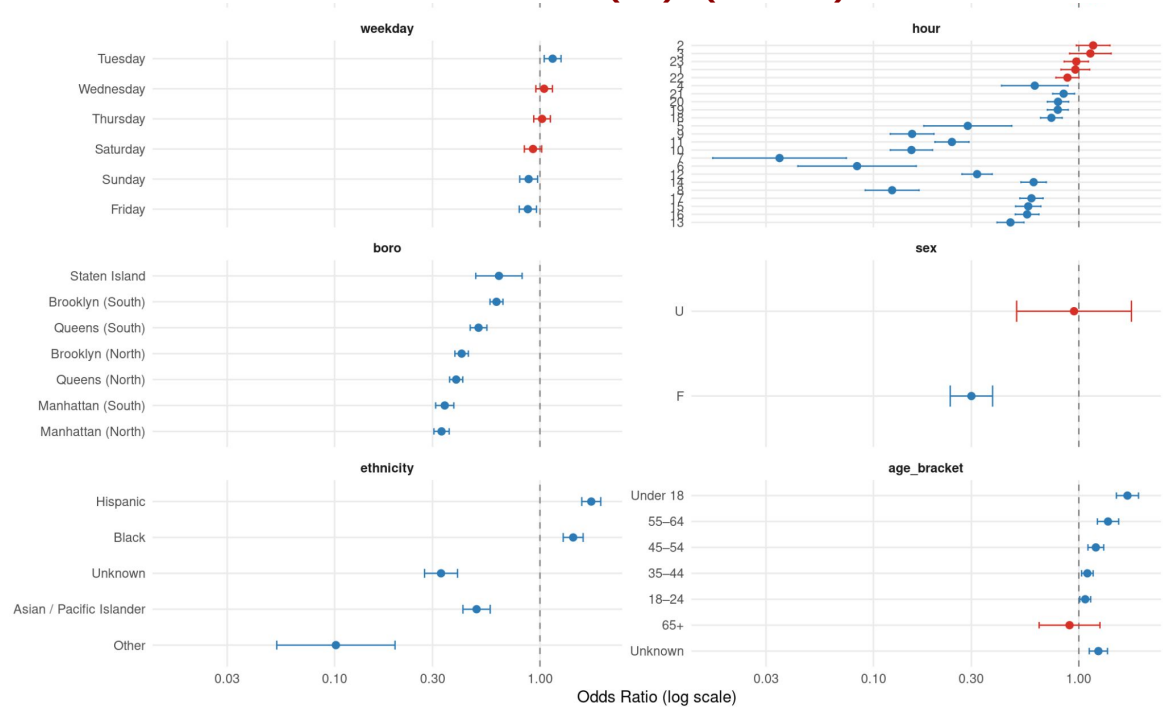
Question: Do signals persist under different model assumptions?

Logistic Regression: Key Effects

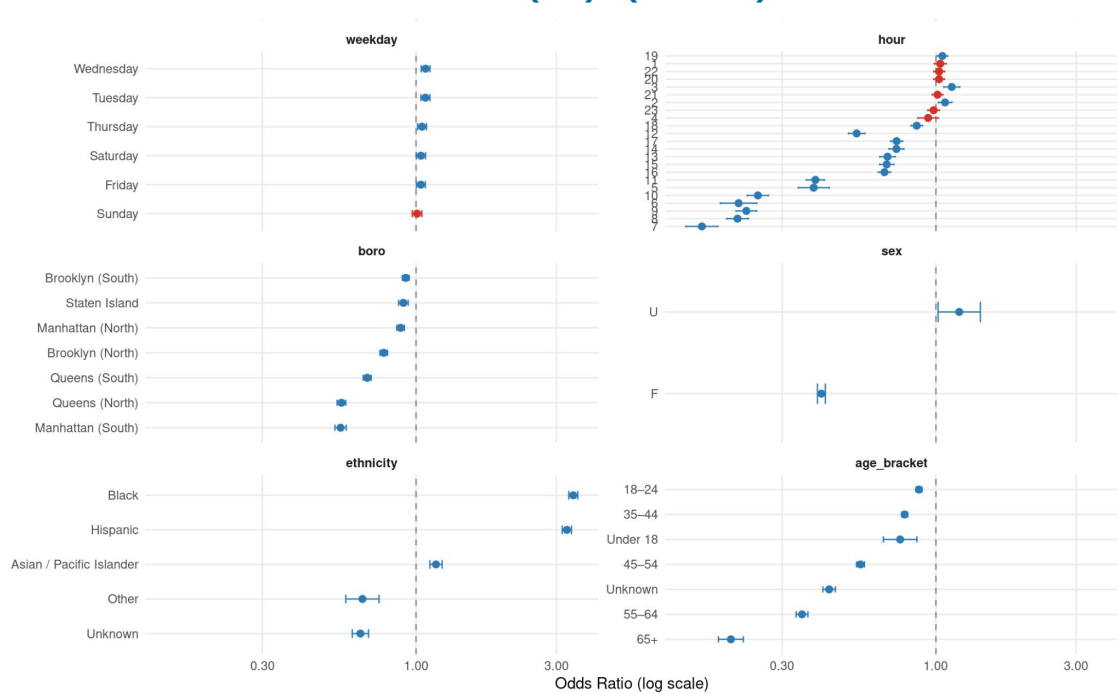
EDA patterns do not disappear

The predictors, whose coefficients have the most impact on the log odds, seem to be hour, boro, sex, ethnicity, and age, like the EDA suggested

2-Wheeled Vehicles (2W) - (Selected) OR Plots



4-Wheeled Vehicles (4W) - (Selected) OR Plots



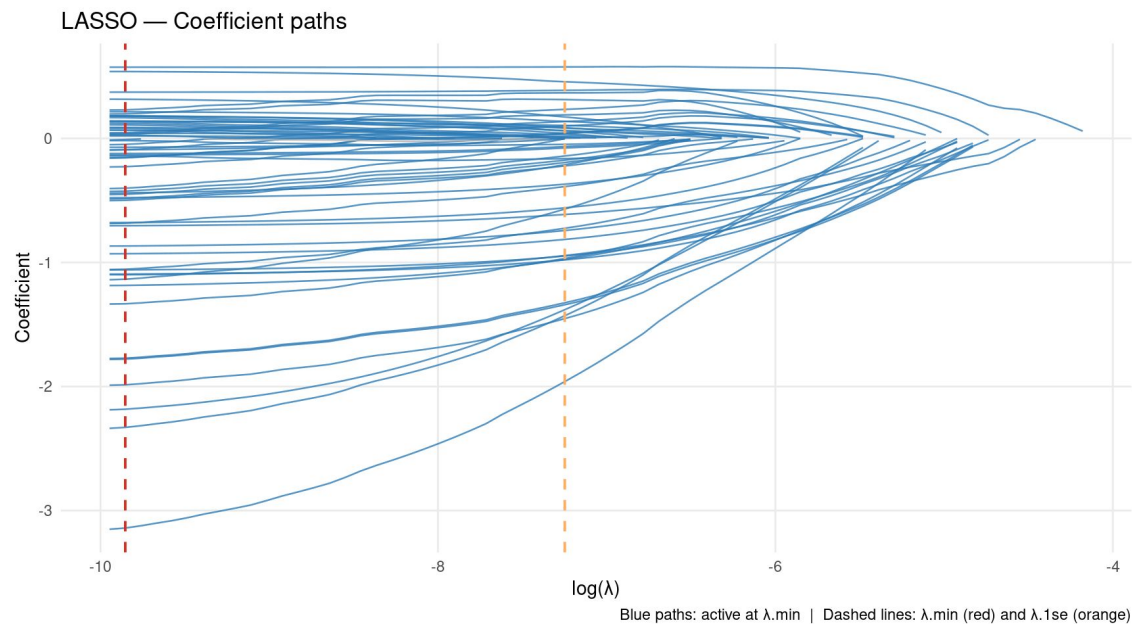
Reference levels: year = 2023, month = 1, weekday = Monday,
hour = 0, boro = Bronx,
sex = M, ethnicity = White, age_bracket = 25-34

LASSO: Shrinkage & Stability

2W Regularization Path

62 → 53 Features

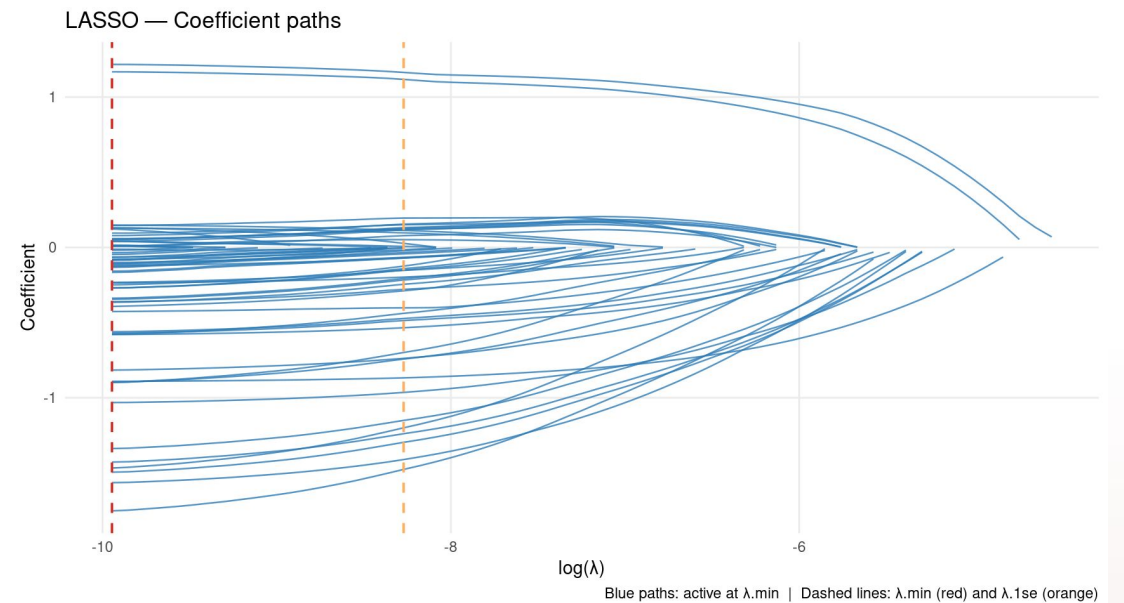
Selection at λ_{1se} (7.11×10^{-4}) reduces model complexity with a **small change** in CV ROC-AUC (0.7394 → 0.7373)



4W Regularization Path

63 → 50 Features

Selection at λ_{1se} compresses parameters by 20% with a **small change** in CV ROC-AUC (CV-AUC 0.7524 → 0.7515)



LASSO: Shows that Signals Persist



Hour 07 Notch

Remains strongly negative under aggressive L1 penalization (OR = 0.141 for 2W, 0.228 for 4W)



Demographics

Black and Hispanic coefficients retain large positive magnitudes, while female remains strongly negative



Geography

Borough-level indicators survive penalization, confirming spatial variation is not noise

2-Wheeled Vehicles (2W) - Example of strong signals

Coefficient	Log Odds → OR (min)	Log Odds → OR (1se)
hour7	-3.1408 → 0.043 **	-1.9603 → 0.141
ethnicityHispanic	0.5745 → 1.776	0.5764 → 1.780
sexF	-1.1848 → 0.306	-0.9757 → 0.377

4-Wheeled Vehicles (4W) - Example of strong signals

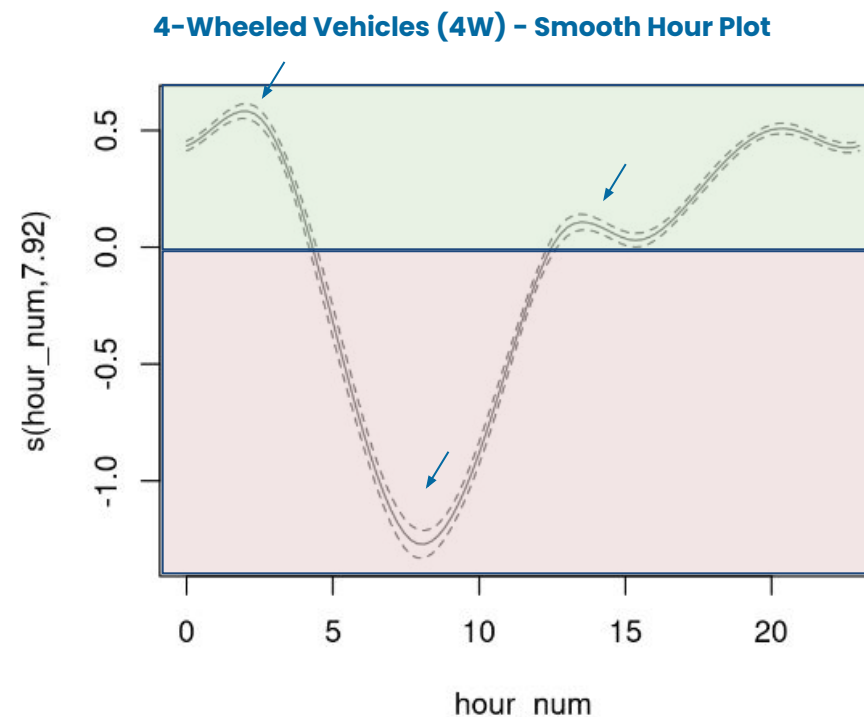
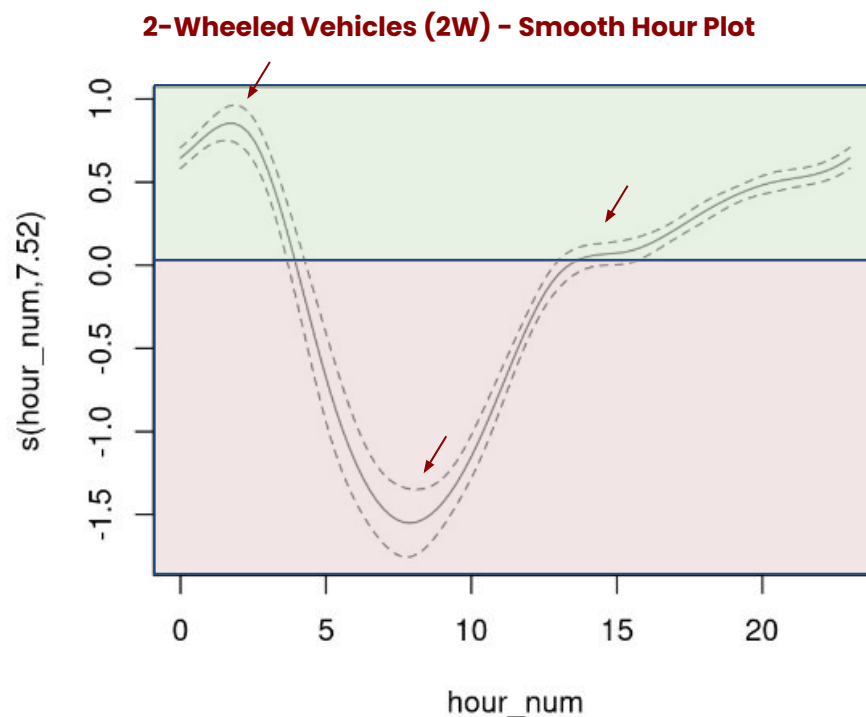
Coefficient	Log Odds → OR (min)	Log Odds → OR (1se)
hour7	-1.7520 → 0.173	-1.4763 → 0.228
ethnicityHispanic	1.1679 → 3.215	1.1174 → 3.057
sexF	-0.8905 → 0.410	-0.8674 → 0.420

Assuming hour0 is the reference category, OR of hour7 = 0.043 means that, holding the other model variables constant, observations occurring at hour 7 have estimated odds of arrest approximately 95.7% lower than the reference hour**

GAM: Confirmed Common Non-Linear Hour Effect

Smooth Diurnal Splines

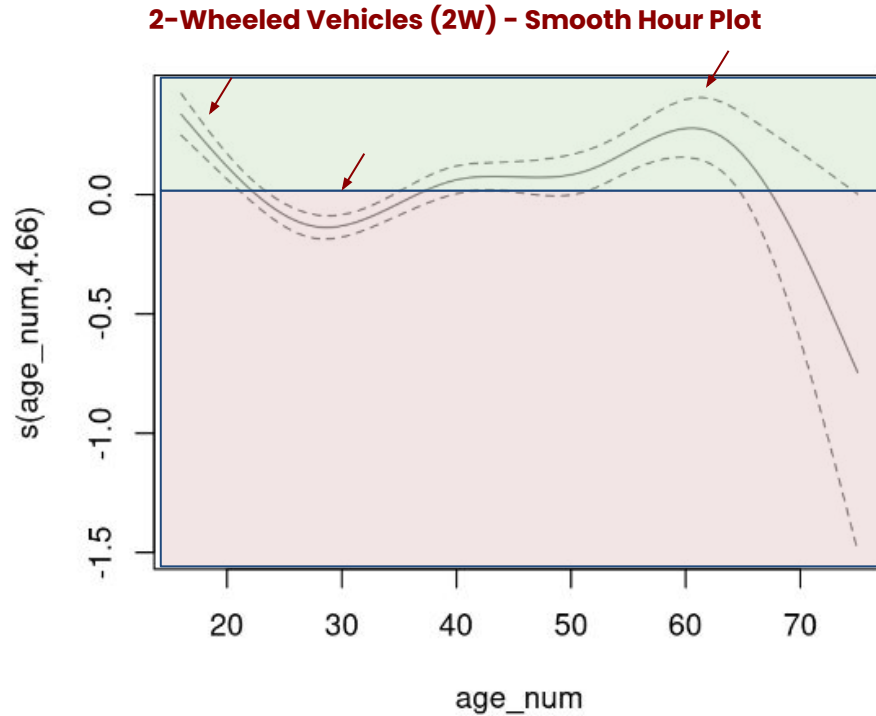
Generalized Additive Models confirm strong non-linearity for hour in both populations (edf = 7.52 for 2W, edf = 7.92 for 4W, $p < 2 \times 10^{-16}$).



Daily Pattern Sequence

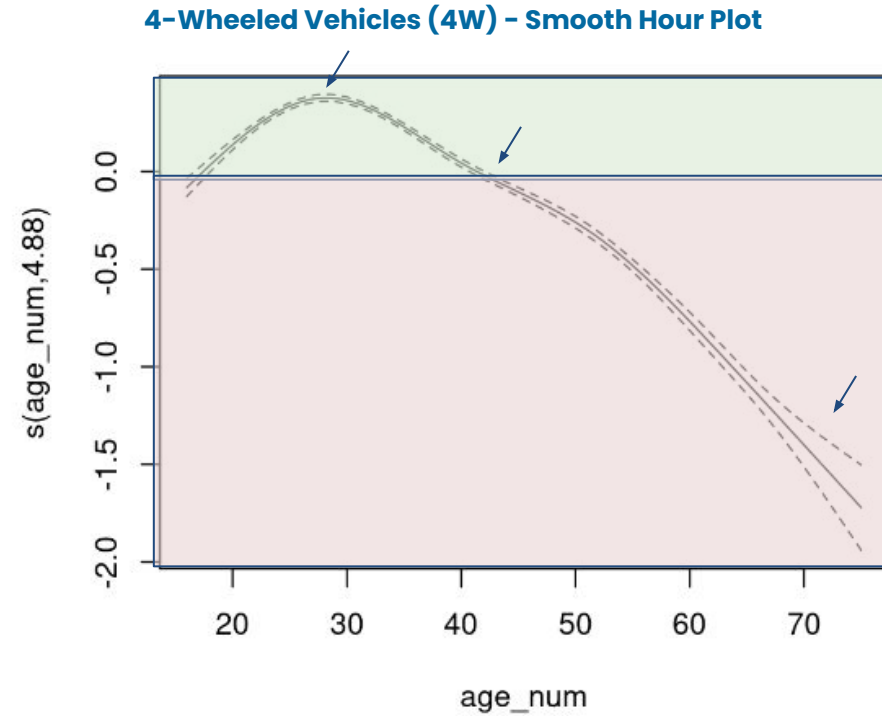
Late night peak → Sharp morning drop (min 07:00–08:00) → Steady afternoon/evening recovery.

GAM: Age Separates the Populations



2W Age Spline (edf = 4.66)

Conditional arrest propensity is elevated among the youngest drivers, and the oldest.



4W Age Spline (edf = 4.88)

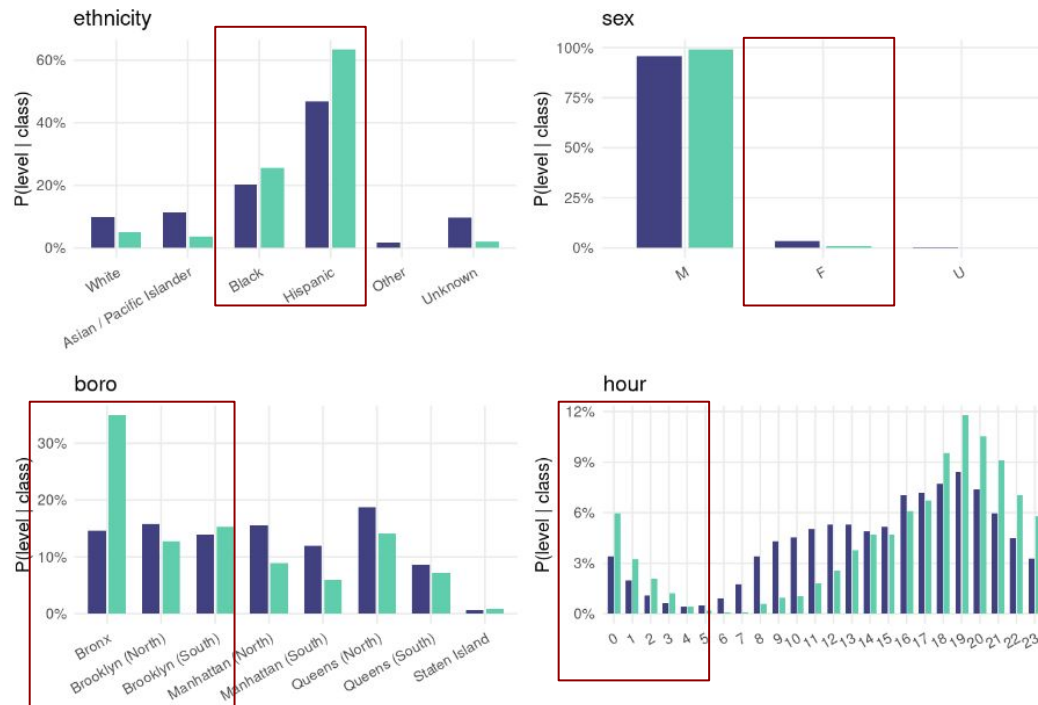
Conditional arrest propensity peaks cleanly around ages 27–30, followed by a decline at older ages.

Naive Bayes: Different Assumptions, Same Class Separators

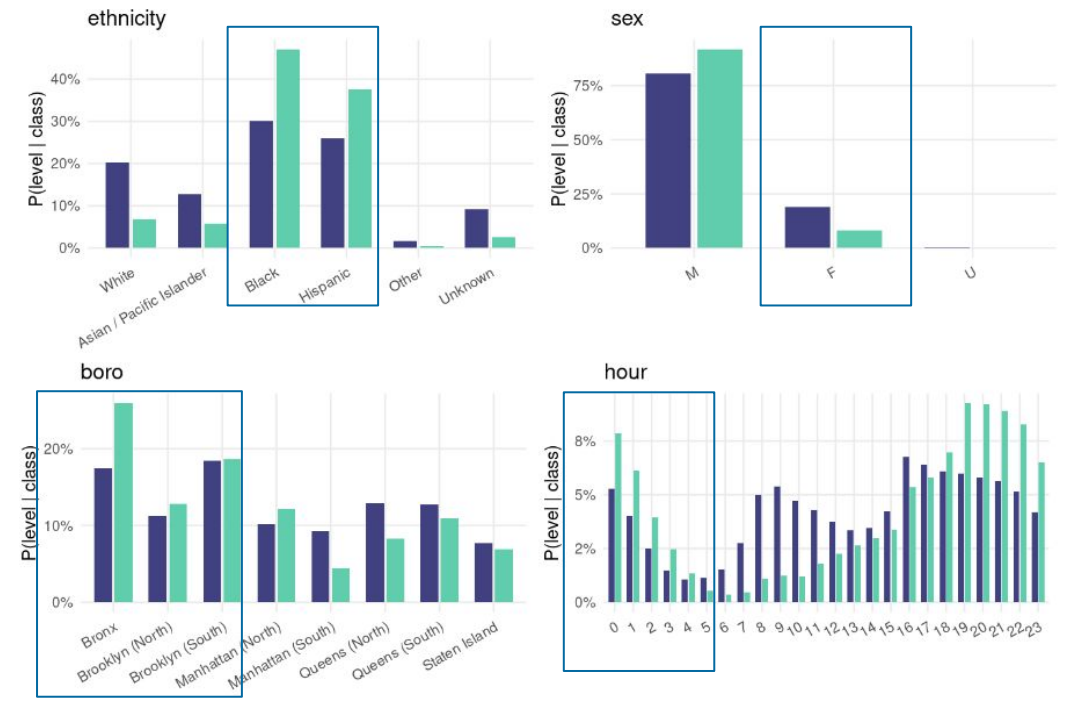
Alternative Probabilistic Modeling

Naive Bayes models $P(X|Y)$ using class-conditional distributions. Despite strong independence assumptions, it isolates the exact same separators.

2-Wheeled Vehicles (2W) - (Selected) Conditional Probabilities



4-Wheeled Vehicles (4W) - (Selected) Conditional Probabilities



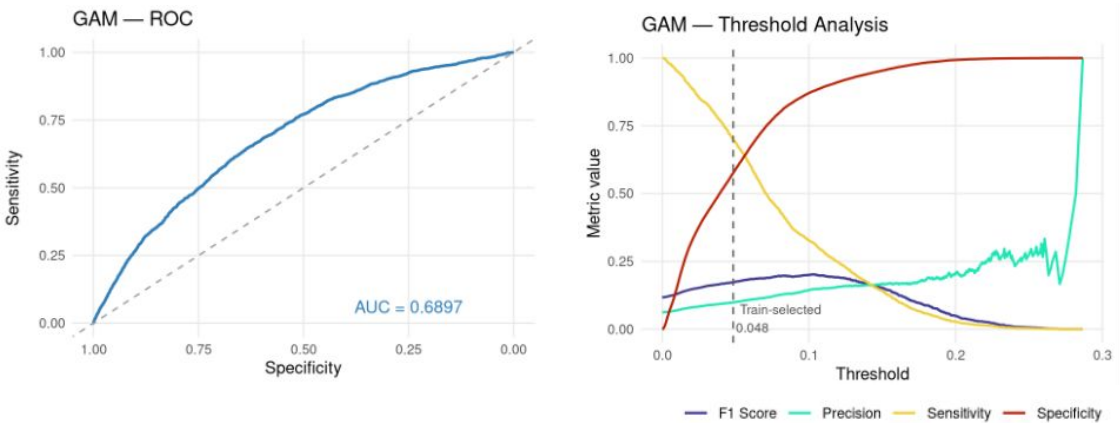
Not arrested Arrested

Class Priors $P(Y=1)$ in %: 2W: 5.38% | 4W: 3.25%

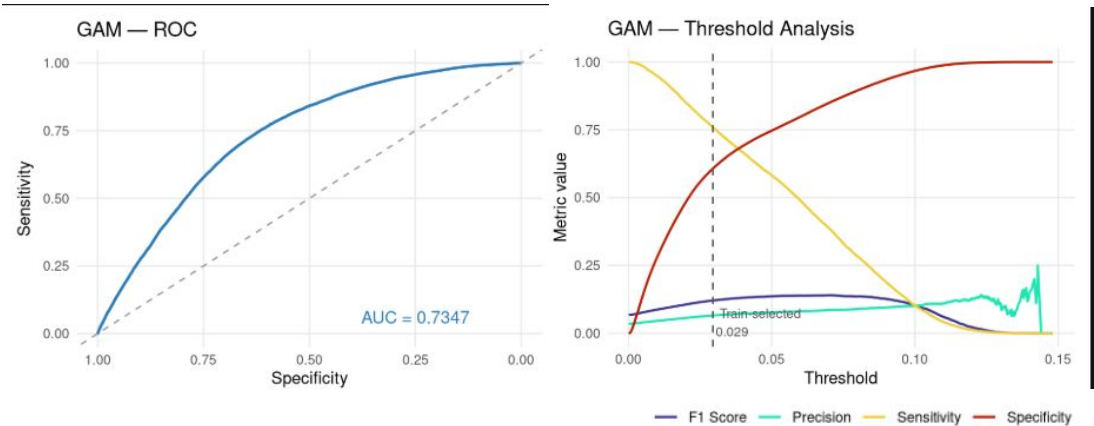
Model Comparison & Predictive Upper Limit



GAM for 2-Wheeled Vehicles (2W) - Eval Plots



GAM for 4-Wheeled Vehicles (4W) - Eval Plots



Model Architecture	2W ROC-AUC	2W PR-AUC	4W ROC-AUC	4W PR-AUC
Logistic Regression	0.6868	0.1207	0.7352	0.0774
LASSO (λ_{1se})	0.6866	0.1203	0.7340	0.0763
GAM	0.6897	0.1247	0.7347	0.0790
Naive Bayes	0.6818	0.1169	0.7322	0.0761

Model Capabilities, Limits and Conclusions

✓ What Models Are Good For

- ✓ Interpreting odds ratios
- ✓ Comparing 2W vs 4W populations
- ✓ Mapping non-linear daily time cycles

⚠ What Models CANNOT Do

- ✗ Reliable individual arrest classification
- ✗ High precision (PR-AUC remains low)
- ✗ High deviance explanation (GAM ~9–10%)

We found some persistent pattern

Hour, borough, and demographics remain significantly associated with arrest after controlling the remaining predictors.

But arrests are still hard to explain

Hour, borough, and demographics remain significantly associated with arrest after controlling the remaining predictors.

Main Conclusion

Recorded NYPD vehicle stops exhibit persistent post-stop arrest disparities, but the data does not establish a causal explanation for those disparities.